

Geometric Origin of Quantum Bounds: Deriving the Uncertainty Principle from Hyperbolic Phase Space Curvature

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Abstract

The Heisenberg Uncertainty Principle (HUP), $\Delta x \Delta p \geq \hbar/2$, is typically derived algebraically from the canonical commutation relation (CCR), $[\hat{x}, \hat{p}] = i\hbar$. We present a new derivation that establishes the HUP as a **geometric necessity** stemming directly from the **Universal Hyperbolic Law (UHL)** [1, 2]. By assuming the generalized momentum $\hat{p}$ is a bounded compositional quantity, the phase space manifold must be hyperbolic with a metric determined by the bounding constant κ . This geometric constraint forces the operator algebra to deviate from the standard CCR. We derive the **Hyperbolic Canonical Commutation Relation (HCCR)**: $[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2)$, and the resulting **Hyperbolic Uncertainty Principle (HUP)**: $\Delta x \Delta p \geq \frac{\hbar}{2}(1 - \kappa^2 \langle \hat{p}^2 \rangle)$. This relation shows that the $\hbar/2$ bound is the natural minimal area element of a curved phase space, and crucially, predicts a **momentum-dependent correction** to the uncertainty limit. This correction links the kinematic bounds of Special Relativity ($\kappa = 1/c$) directly to the observational bounds of Quantum Mechanics ($\hbar$), predicting that the quantum uncertainty bound vanishes for states tightly constrained to the physical momentum limit.

Keywords: Uncertainty Principle; Hyperbolic Geometry; Phase Space; Canonical Commutation Relation; Universal Hyperbolic Law; Quantum Foundations.

1 Introduction

The Universal Hyperbolic Law (UHL) [1, 2] posits that any bounded, compositional one-dimensional quantity $u \in (-1/\kappa, 1/\kappa)$ must be linearizable by the **rapidity** coordinate $r = \frac{1}{\kappa} \operatorname{artanh}(\kappa u)$. This framework successfully derives the Einstein velocity addition law ($\kappa = 1/c$) [1] and the group action for bounded dynamical systems [2]. We extend the UHL to the quantum domain by arguing that if the dynamics of a quantity are constrained by a hyperbolic boundary, its conjugate pair of observables must reflect this underlying hyperbolic geometry.

2 The Hyperbolic Phase Space Axioms

2.1 Axiom 1: Bounded Momentum

The generalized momentum $\hat{p}$ is a bounded, compositional observable whose eigenvalues must reside in the interval $I_\kappa = (-1/\kappa, 1/\kappa)$. This axiom asserts that the geometry of the physical phase space is **hyperbolic**, determined by the maximum scale κ .

2.2 Axiom 2: Rapidity Commutation

The rapidity operator $\hat{r} = \frac{1}{\kappa} \operatorname{artanh}(\kappa\hat{p})$ linearizes the curved hyperbolic manifold. We assume $\hat{r}$ and its conjugate position $\hat{x}$ obey the standard, flat-space CCR:

$$[\hat{x}, \hat{r}] = i\hbar\hat{I} \quad (1)$$

This transfers the fundamental quantum scale ($\hbar$) to the geometrically flat coordinate, ensuring the $\hat{x} - \hat{p}$ non-commutation arises purely from the hyperbolic map.

3 Derivation of the Hyperbolic Commutation Relation (HCCR)

We calculate $[\hat{x}, \hat{p}]$ using the functional identity $[\hat{A}, f(\hat{B})] = [\hat{A}, \hat{B}]f'(\hat{B})$, where $\hat{p} = f(\hat{r}) = \frac{1}{\kappa} \tanh(\kappa\hat{r})$.

$$[\hat{x}, \hat{p}] = [\hat{x}, \hat{r}] \frac{d}{d\hat{r}} \left(\frac{1}{\kappa} \tanh(\kappa\hat{r}) \right) = i\hbar\hat{I} \cdot \operatorname{sech}^2(\kappa\hat{r}) \quad (2)$$

Using the identity $\operatorname{sech}^2(\kappa\hat{r}) = 1 - \tanh^2(\kappa\hat{r}) = 1 - \kappa^2\hat{p}^2$, the **Hyperbolic Canonical Commutation Relation (HCCR)** is:

$$[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2\hat{p}^2) \quad (3)$$

4 The Hyperbolic Uncertainty Principle (HUP)

Applying the Robertson-Schrödinger uncertainty relation, $\Delta x \Delta p \geq \frac{1}{2} |\langle [\hat{x}, \hat{p}] \rangle|$:

$$\Delta x \Delta p \geq \frac{\hbar}{2} (1 - \kappa^2 \langle \hat{p}^2 \rangle) \quad (4)$$

4.1 Testable Prediction of Vanishing Uncertainty

The bound is dependent on the state's average squared momentum, $\langle \hat{p}^2 \rangle = (\Delta p)^2 + \langle p \rangle^2$. A crucial, testable prediction is that for states localized near the boundary ($\langle \hat{p}^2 \rangle \rightarrow 1/\kappa^2$), the quantum uncertainty bound itself **vanishes** ($\Delta x \Delta p \rightarrow 0$). This is a fundamental deviation from standard Quantum Field Theory and a direct consequence of the hyperbolic geometry imposing a classical-like constraint at the physical limit.

5 Conclusion

This derivation establishes a unified geometric foundation for both bounded kinematics and quantum observability under the UHL. The resulting HCCR is the foundational quantum axiom for the next phase of the research program: the **Dirac Transposition**.

This algebraic form differs from other proposed hyperbolic CCRs, such as the one in Kiukas & Werner (2006) [math-ph/0511044], which relies on split-complex numbers and lacks the explicit momentum-dependent correction term derived here from the tanh rapidity map.

References

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